

LAAYOUNE, Western Sahara

WMO: 600330

Lat: 27.152N

Lon: 13.219W

Elev: 63

StdP: 100.57

Time Zone: 1.00 (E01)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	11.7	-8.1	1.9	20.9	-5.8	2.3	21.2	14.0	18.9	12.2	19.2	3.2	120	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	36.1	20.7	33.8	20.6	31.2	20.3	23.7	31.0	22.9	29.9	22.3	28.4	6.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.8	26.3	21.1	15.9	24.6	20.2	15.0	23.8	71.3	31.0	67.9	30.2	65.6	28.7	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	12.1	11.2	DB	7.6	42.0	1.9	2.9	6.2	44.1	5.1	45.8	4.1	47.4	2.7	49.5	(4)
(5)			WB	4.0	24.9	1.4	1.0	2.9	25.6	2.1	26.2	1.3	26.7	0.2	27.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	17.3	18.0	19.8	20.2	21.2	22.8	24.7	25.7	25.0	23.8	20.8	18.5	(6)	
	DBStd	3.52	2.00	2.31	2.64	2.54	2.16	1.75	2.06	1.96	1.72	2.33	2.65	1.84	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	115	44	31	11	4	1	0	0	0	0	0	4	21	(9)	
	CDD10.0	4197	227	224	303	306	347	383	455	488	451	427	324	263	(10)	
	CDD18.3	1270	13	22	55	59	88	133	197	229	201	169	79	26	(11)	
	CDH23.3	8028	101	151	380	359	423	707	1273	1659	1348	1044	443	140	(12)	
(13)	CDH26.7	2768	13	37	152	147	138	171	447	682	467	356	139	19	(13)	
(14)	Wind	WSAvg	6.3	5.2	5.6	5.8	6.4	6.7	7.4	8.9	8.3	6.3	4.9	5.1	(14)	
(15) Precipitation	PrecAvg	55	13	9	6	1	1	0	0	1	1	3	6	15	(15)	
	PrecMax	128	61	60	21	7	4	1	1	7	6	15	26	58	(16)	
	PrecMin	12	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	35	18	14	6	2	1	0	0	2	1	4	8	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	30.1	34.9	37.1	38.1	35.0	39.0	38.2	36.2	37.0	33.1	28.9	(19)	
		MCWB	13.9	14.5	16.6	16.9	19.2	21.6	23.5	24.2	22.0	20.7	17.3	14.2	(20)	
	2%	DB	26.0	27.2	30.9	30.2	30.1	30.2	33.1	34.9	32.8	33.0	30.2	26.8	(21)	
		MCWB	12.6	14.5	16.4	16.8	18.6	20.7	22.3	23.0	22.0	19.2	16.5	13.8	(22)	
	5%	DB	24.1	25.0	27.3	26.8	27.0	28.1	30.2	32.0	30.9	30.1	28.0	24.9	(23)	
		MCWB	12.8	14.7	16.1	17.2	18.6	20.0	21.4	22.3	21.3	20.1	16.9	14.0	(24)	
	10%	DB	22.1	22.8	24.2	24.2	25.1	26.9	28.8	30.0	29.1	28.0	25.8	23.1	(25)	
		MCWB	13.4	14.9	16.1	16.8	17.9	19.4	20.7	21.7	20.8	20.0	17.2	14.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	18.9	19.9	20.0	21.2	22.7	24.3	25.6	24.3	23.6	21.4	20.0	(27)	
		MCDB	22.2	23.5	27.2	27.8	30.3	30.6	33.1	33.5	30.6	29.7	26.9	23.1	(28)	
	2%	WB	16.8	17.7	18.5	18.8	20.0	21.5	23.0	24.0	23.1	22.5	20.5	18.6	(29)	
		MCDB	20.6	22.4	24.6	25.2	27.2	28.1	31.6	31.3	29.1	27.7	25.0	21.5	(30)	
	5%	WB	15.9	16.9	17.7	18.1	19.1	20.7	22.1	23.0	22.4	21.7	19.6	17.5	(31)	
		MCDB	19.7	21.5	22.9	24.0	25.0	26.6	28.9	30.5	28.0	26.2	23.8	20.6	(32)	
	10%	WB	15.2	16.1	17.1	17.3	18.4	19.9	21.2	22.3	21.6	21.1	18.8	16.6	(33)	
		MCDB	19.0	20.2	22.0	22.6	23.7	25.4	27.1	28.5	26.8	25.3	22.7	19.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	10.5	9.4	9.0	9.1	9.5	10.1	9.9	9.7	9.7	10.0	(35)	
		MCDBR	13.2	13.9	15.3	13.5	12.3	11.3	12.1	13.2	11.9	12.9	12.7	12.5	(36)	
	5% WB	MCWBR	6.0	5.9	5.4	4.6	4.2	3.7	3.7	3.9	3.6	4.2	5.0	5.7	(37)	
		MCWBR	9.7	10.7	11.9	11.2	10.8	10.9	11.4	12.0	10.5	10.1	9.8	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.396	0.428	0.466	0.461	0.451	0.518	0.651	0.640	0.535	0.460	0.407	0.406	(40)	
		taud	2.155	2.058	1.950	1.998	2.055	1.884	1.611	1.655	1.886	2.059	2.185	2.140	(41)	
	Ebn at Noon		834	836	827	843	848	788	688	692	756	794	816	801	(42)	
		Edh at Noon	131	156	184	180	170	201	263	251	194	154	126	128	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.04	4.94	6.10	6.94	7.40	7.49	7.18	6.72	6.06	5.06	4.15	3.65	(44)	
	RadStd		0.20	0.24	0.22	0.26	0.25	0.21	0.22	0.21	0.29	0.21	0.17	0.19	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page